Journal of Alloys and Compounds 472 (2009) 241–246

Contents lists available at ScienceDirect

Journal of Alloys and Compounds

journal homepage: www.elsevier.com/locate/jallcom

Oxidation of Ti–6Al–4V alloy

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ARTICLE INFO

Article history:
Received 17 February 2008
Received in revised form 6 April 2008
Accepted 9 April 2008
Available online 27 May 2008

Keywords:
Metals and alloys
Diffusion
Oxidation

ABSTRACT

Air oxidation behaviour of a Ti–6Al–4V alloy was examined over the temperature range of 600–800 °C for different time intervals ranging between 0.5 and 72 h. In addition to the determination of the oxidation kinetics by weight gain measurements, oxygen diffusion zone (ODZ) depth and hardness measurements were conducted on the cross-sections of the oxidised alloys to investigate the diffusion kinetics of oxygen into the substrate. The rate of oxidation evaluated according to the weight gain measurements fitted parabolic kinetics between 600 and 700 °C and linear kinetics above 700 °C by yielding activation energies of 276 and 191 kJ/mol, respectively. Analysis of the ODZ depth and hardness measurements resulted in an activation energy of 202 kJ/mol for oxygen diffusion in Ti–6Al–4V alloy. Finally, the kinetics of oxidation and oxygen diffusion were evaluated by the analysis of the experimental data with respect to temperature-compensated time parameter.

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1. Introduction

Titanium and its alloys are widely used in a variety of applications in aerospace, automotive, chemical and biomedical industries owing to the superior combination of mechanical properties with low density and good corrosion resistance. High affinity for oxygen limits their use at elevated temperatures. Thus, they suffer from extensive scaling when exposed to oxidising environment mostly at relatively high temperatures [1–3].

Exposure of titanium and its alloys to any oxygen containing atmosphere at elevated temperatures leads to formation of an oxide layer (OL) on the surface with an oxygen diffusion zone (ODZ) beneath it. Rutile and anatase are the most common modifications of TiO_2 that form as an OL. Irreversible anatase to rutile phase transition takes place within the OL at about 700 °C [4]. Dissolution of oxygen in the underlying substrate as a result of inward diffusion of oxygen causes development of hard and brittle ODZ. Since it leads a loss of tensile ductility and of fatigue resistance, ODZ formation is critical to the life expectancy of titanium alloys, for example when used in aero-gas turbine engines [5].

Addition of aluminum considerably enhances the oxidation behaviour of titanium by forming thermodynamically stable, protective α-Al_2O_3 phase. If aluminum content is sufficient, OL will be composed of a mixture of TiO_2 and α-Al_2O_3 [2,6]. On the other hand aluminum content is limited to maximum 6 wt.% for conventional titanium alloys due to its detrimental effect on ductility [2]. For Al–Ti alloys, Shida and Anada [7] classified the alloying elements into three groups according to their effects on oxidation resistance as: (i) detrimental; V, Cr, Mn, Pd, Pt, Cu; (ii) neutral; Y, Zr, Hf, Ta, Fe, Co, Ni, Ag, Au, Sn, O; and (iii) beneficial; Nb, Mo, W, Si, Al, C, B. Among the beneficial alloying elements Nb is very attractive, because it also provides an improvement in creep resistance and room temperature toughness. Jiang et al. [8] reported that Nb may lead a reduction in the oxidation resistance when the amount of Nb is too high and forms $TiNb_2O_7$ or $AlNbO_4$ within the OL.

The high affinity of titanium to oxygen causes formation of a very thin OL on the surface even in air at room temperature. Above 200 °C the thickness of the OL significantly increases. ODZ develops above 400 °C. At temperatures above 600 °C, increase of oxidation rate causes formation of a thicker but defective OL leading to easy penetration of oxygen into the substrate. Oxidation at elevated temperatures (mostly above 800 °C) remarkably degrades the high temperature performance of titanium and its alloys not only as the result of thinning of the material by detachment of thick OL, but also due to formation of wider ODZ [1–3].

Recently, the high temperature oxidation manifested itself as a very promising surface modification technique for titanium and its alloys to provide remarkable protection against friction and wear. This surface modification technique, which benefits from the oxidation behaviour of titanium, is known as thermal oxidation. For titanium and its alloys, the ultimate goal of thermal oxidation is to create relatively thin, mechanically stable OL supported by the ODZ for successful surface protection at room temperature [8–19].

Therefore, there is a great interest in analysing the oxidation behaviour of titanium and its alloys not only for interpreting the degree of degradation at high service temperatures, but also for

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doi:10.1016/j.jallcom.2008.04.024

enhancing room temperature friction and wear properties through thermal oxidation. In this respect, the objective of this work was to study the oxidation and oxygen diffusion kinetics of the most popular Ti–6Al–4V alloy, which covers about 50% of total titanium production. In the literature, reports on oxidation behaviour of Ti–6Al–4V alloys are scarce [20,21]. Additionally, an attempt has been made in the present study for Ti–6Al–4V alloy to predict the extent of oxidation by utilising a temperature-compensated time parameter in order to estimate the service life at elevated temperatures or room temperature wear performance after thermal oxidation. Previously, an oxidation model has been proposed for titanium alloys by the aid of oxidation kinetics to predict the life of the components to prevent catastrophic failures during service [5].

2. Experimental

In this study, Ti–6Al–4V was subjected to oxidation at various temperatures in between 600 and 800 °C, for different time intervals ranging between 0.5 and 72 h. Before oxidation, 5 mm thick cylindrical samples cut from the 8 mm diameter bar were successively ground on 240–1200 mesh SiC abrasive papers. Final polishing was made with 1 μm Al_2O_3 slurry abrasive to achieve a mirror-like surface finish. Later they were cleaned in acetone and dried in hot air.

Isothermal oxidation studies were carried out in a laboratory type air circulated furnace. Weights of the samples were measured by an electronic balance having accuracy of ±0.1 mg, before and after oxidation. The weight gain values of the three samples were averaged for each oxidation condition.

Microstructural examinations of the oxidised samples were conducted by X-ray diffraction (XRD) analysis and light optical microscope (LOM) survey. XRD analysis was carried out under Cu Kα radiation with the incidence beam angle of 2°. The diffraction angle range was between 10° and 90° with a step increment of 0.02° and a count time of 1 s. LOM survey was conducted on cross-sections of the samples, which were prepared by standard metallographic technique. After grinding and polishing, microstructural evolution was elucidated by etching with 2% HF solution. Additionally, hardness measurements were carried out on the cross-sections of the polished samples with a Vickers indenter under the indentation load of 5 g by utilising Fischer HP 100 XY-PROG ultramicrohardness tester.

3. Results and discussion

3.1. Analysis of weight gain measurements

Fig. 1 presents the variation of normalised weight gain ($\Delta W/A$) data with respect to oxidation time (t). Normalised weight gain was calculated in the unit of mg/cm^2 by dividing the weight gain values of the oxidised samples (ΔW) to their total surface areas (A).

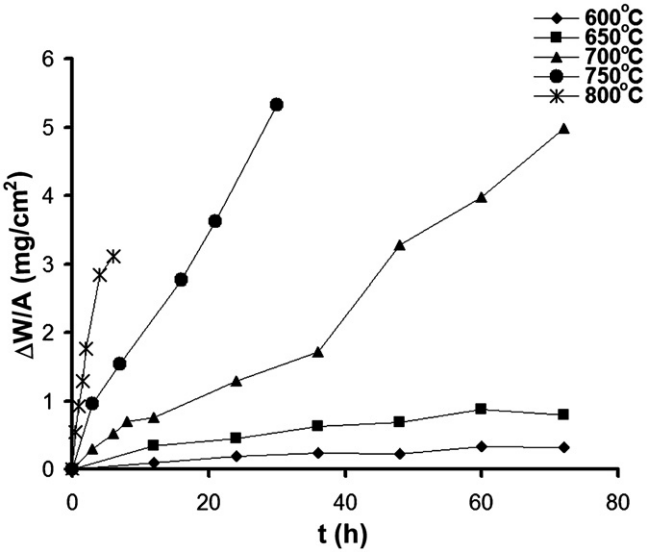


Fig. 1. The variation of normalised weight gain ($\Delta W/A$) with respect to oxidation time (t) and temperature.

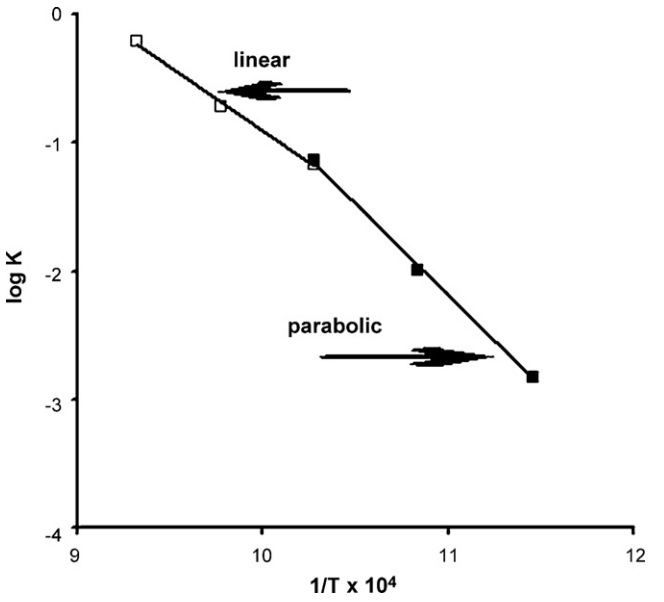


Fig. 2. The Arrhenius plot of the rate constant (K) values calculated by utilizing normalised weight gain ($\Delta W/A$) values presented in Fig. 1.

Fig. 1 shows that increase of oxidation temperature accelerated the weight gain, which obeyed parabolic and linear oxidation kinetics at 600–650 and 750–800 °C, respectively. Weight gain values of 750 and 800 °C for the oxidation times longer than 30 and 6 h, respectively, are not included in Fig. 1, due to the scaling of OL. At 700 °C parabolic oxidation was dominant up to 36 h of oxidation, and prolonged oxidation yielded linear oxidation kinetics.

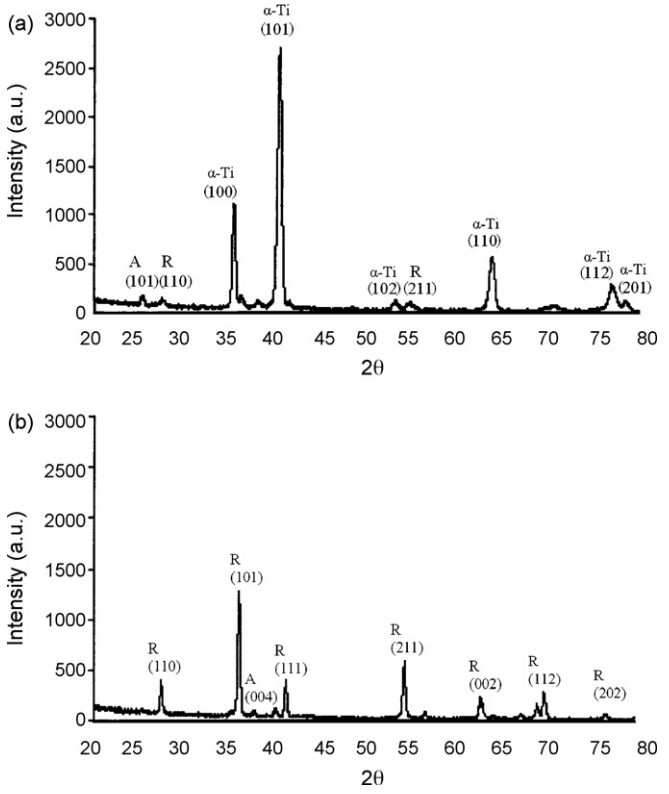


Fig. 3. XRD patterns of the samples oxidised at (a) 600 °C for 12 h and (b) 650 °C for 48 h (R: rutile; A: anatase).

Conventionally, analysis of oxidation kinetics and determination of the oxidation rate is successfully carried out by fitting the weight gain data into the following equation [3]:

$$\left(\frac{\Delta W}{A}\right) = Kt^{1/n}, \quad (1)$$

where K is the rate constant and n is the reaction index. At a constant temperature, the kinetic model is linear if $n = 1$ and parabolic if $n = 2$. Accordingly, regression analysis of the weight gain data would yield the K value through calculation of the slope of the best fit line of $(\Delta W/A)$ vs. t plot. Temperature dependence of K can be quantified by an Arrhenius type equation [3]:

$$K = K_0 \exp\left(\frac{-Q}{RT}\right), \quad (2)$$

where K_0 is the frequency factor, Q is the activation energy, R is the gas constant (8.3143 J/(mol K)) and T is reaction temperature (K). The value of Q can be calculated by plotting $\log K$ with respect to $1/T$ in which the slope of the best fit line is equal to $Q/2.303R$.

In Fig. 2, the K values, which were obtained after the regression analysis of the $\Delta W/A$ vs. t curves (Fig. 1), are presented as an Arrhenius plot for the examined Ti–6Al–4V alloy. For the oxidation temperature of 700 °C, two K values corresponding to parabolic and linear oxidation stages were calculated. The slope of the best fit line obtained from the "$\log K$ vs. $1/T$" plot yielded the activation energy of oxidation (Q_{ox}) as 276 kJ/mol for the parabolic and 191 kJ/mol for the linear oxidation stages. It should be noted that these Q_{ox} values were calculated for the maximum oxidation time of 72 h.

Previous oxidation kinetics studies of Ti–6Al–4V alloy reported Q_{ox} values as 192 kJ/mol [20] and 267 kJ/mol [21] for oxidation temperatures ranging between 600–700 °C (for exposure times up to 300 h) and 650–850 °C (for exposure times up to 100 h), respectively. When compared with the results of this study, it is suggested that oxidation times as long as 300 h at 600–700 °C dominated linear oxidation kinetics and therefore yielded Q_{ox} value of 192 kJ/mol. On the other hand, the Q_{ox} value of 267 kJ/mol calculated for oxidation temperatures between 650 and 850 °C for oxidation times shorter than 100 h, can be associated with the parabolic oxidation stage of this study.

3.2. Analysis of microstructural examinations

Representative XRD patterns of the oxidised Ti–6Al–4V alloys are given in Fig. 3. As evidenced, the surfaces were mainly covered by anatase and rutile modification of TiO_2. After oxidation at relatively low temperatures for short times, i.e. at 600 °C for 12 h (Fig. 3a), α-Ti peaks also appeared on the XRD pattern. This was basically due to the penetration of X-ray beyond the OL through

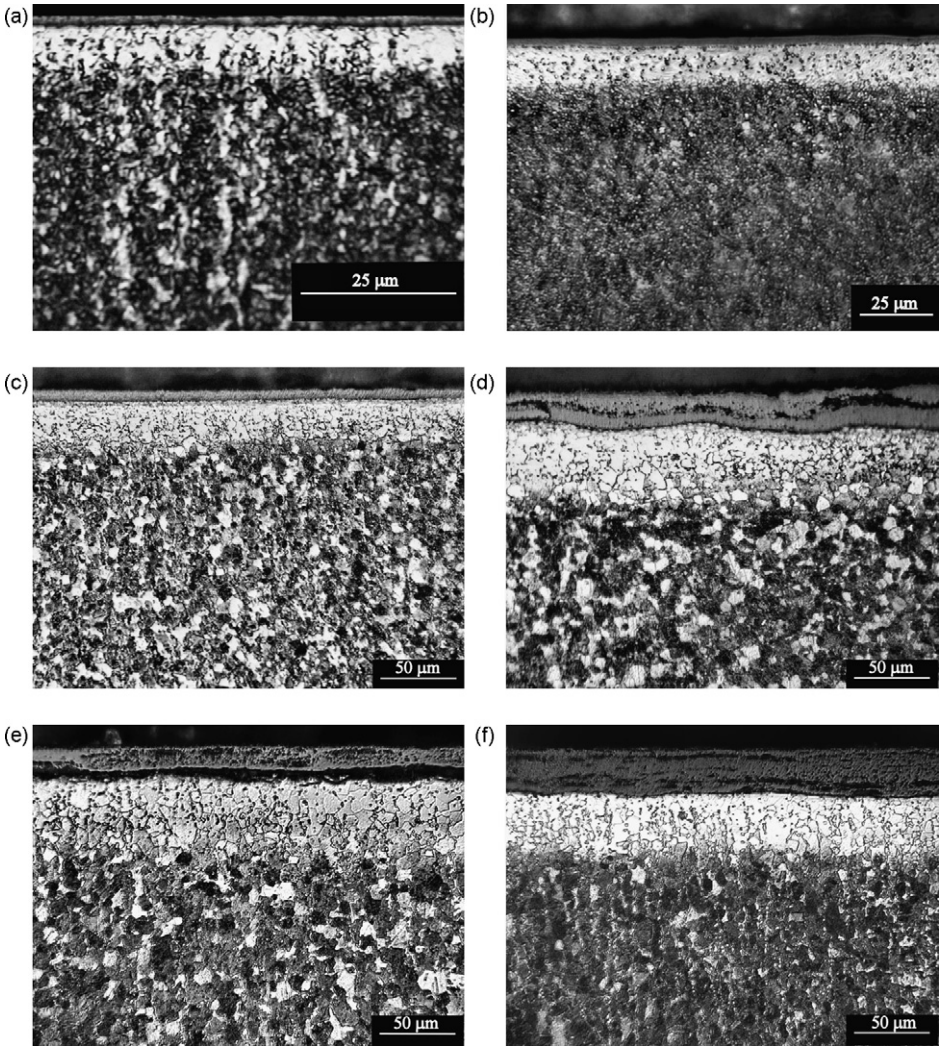


Fig. 4. Cross-sectional LOM micrographs of the samples oxidised at (a) 600 °C for 60 h, (b) 650 °C for 60 h, (c) 700 °C for 24 h, (d) 700 °C for 72 h, (e) 750 °C for 30 h and (f) 800 °C for 6 h.

the substrate. When compared to unoxidised state [17,18], the α-Ti peaks were shifted slightly left from their original positions upon oxidation, which was probably caused by the increase of the lattice parameters due to the dissolution of oxygen in the ODZ. It should be noted that higher oxidation temperatures and longer oxidation times, i.e. at 650 °C for 48 h (Fig. 3b) dominated rutile modification of TiO_2. Especially above oxidation temperature of 650 °C, anatase peak intensities were greatly reduced and α-Ti peaks disappeared on the XRD patterns. Eventhough small amount of TiN_xO_y having face-centered cubic crystal structure has been detected on the XRD pattern of an oxidised Ti–6Al–4V alloy previously [22], presence of oxidation products other than anatase and rutile modifications of TiO_2 were not identified in the present study.

Fig. 4 presents the cross-sectional LOM images of the oxidised samples. OL exhibited two different morphologies (dense and porous) depending on the oxidation temperature and time. As seen in Fig. 4a–c, relatively thin and dense OL exhibited excellent adherence to the substrate. On the other hand, thick and porous OL (Fig. 4d–f) has multilayered structure with limited attachment to the substrate. According to Du et al. [21], high stresses developed due to the thermal expansion coefficient mismatch between the OL and the substrate limits the adherence of the thick OL to the substrate. Yu et al. [23] declared increase of oxidation rate at 700 °C after 20 h due to the failure of the OL. Coddet et al. [24], who studied the oxidation of commercially pure titanium in temperature range of 500–700 °C, claimed that the adhesion strength of the OL was relatively high for temperatures below 650 °C, while it tended to be zero at 700 °C.

The deviation in the morphology of the OL with respect to the oxidation temperature and time can be associated with the oxidation mechanisms mentioned in Section 3.1. Dense OL (Fig. 4a–c) formed at oxidation temperatures (600–700 °C), where parabolic oxidation kinetics was obeyed. Elevated temperatures (700–800 °C), where linear oxidation kinetics was dominant, promoted formation of relatively thick and crumbly OL exhibiting multilayered porous structure (Fig. 4d–f). It is, herein, important to point out that, 700 °C dominated parabolic kinetics by forming dense OL (Fig. 4c) at oxidation times shorter than 36 h and conformed linear kinetics by developing porous OL having multilayered structure (Fig. 4d) at oxidation times longer than 36 h (Fig. 1). The number of layers within the OL tended to increase upon its growth at higher oxidation temperatures and/or longer oxidation times.

According to Du et al. [21] formation of multilayered OL on the surface of Ti–6Al–V alloy is due to alternate growth of Al_2O_3 and TiO_2 layers by outward diffusion of aluminum and inward diffusion of oxygen. At the early stages of oxidation, Al_2O_3 nucleates on the surface together with TiO_2. At extended oxidation times, Al_2O_3 grows laterally and covers TiO_2. In this case, the outer most section of the OL (gas/oxide interface) is Al_2O_3, while the inner section of the OL (oxide/substrate interface) is TiO_2. Until reaching a critical thickness, OL maintains a good contact with the substrate. Once the critical thickness is exceeded crack develops at the interface. Eventhough their peaks were not appeared on the XRD patterns of this study (Fig. 3), Al_2O_3 was detected after oxidation time of 100 h, by Du et al. [21].

As can be seen in Fig. 4, growth of OL on the surface of Ti–6Al–4V alloy was accompanied by development of ODZ due to the diffusion of oxygen beyond the OL. Since oxygen is α-Ti stabilizer [2] and peaks of β-Ti did not appear on the XRD pattern (Fig. 3a), it is suggested that ODZ is α-Ti solid solution. Fig. 5 quantifies the depth of the ODZ (X_{ODZ}) measured from the cross-sections of the etched samples (Fig. 4) with respect to oxidation time and temperature. Longer times and/or higher temperatures resulted with deeper penetration of ODZ. Nevertheless, growth of ODZ obeyed parabolic kinetics at temperature range between 600 and 800 °C.

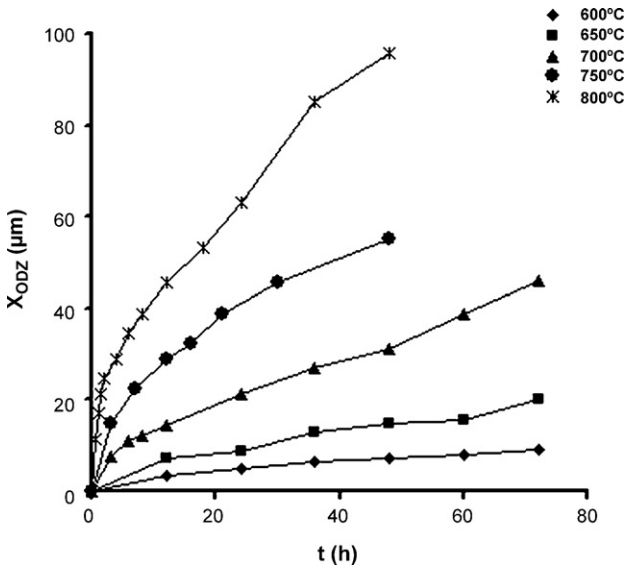


Fig. 5. The variation of depth of ODZ (X_{ODZ}) with respect to oxidation time (t) and temperature.

According to the X_{ODZ} values (Fig. 5), activation energy of ODZ (Q_{ODZ}) was calculated by utilising Eqs. (1) and (2) in similar fashion with Section 3.1. For this purpose, $\Delta W/A$ of Eq. (1) was replaced with X_{ODZ}. K values, which were obtained after the regression analysis of the X_{ODZ} vs. t curves (Fig. 5) are presented in Fig. 6 with respect to reciprocal of oxidation temperature. The slope of the best fit line of Fig. 6 yielded Q_{ODZ} value of 202 kJ/mol.

3.3. Analysis of hardness measurements

The results of the cross-sectional hardness measurements conducted on the samples oxidised at 600 and 650 °C are presented in Fig. 6. Since ODZ was shallow at oxidation times shorter than 60 h, successful hardness measurements were carried out only on

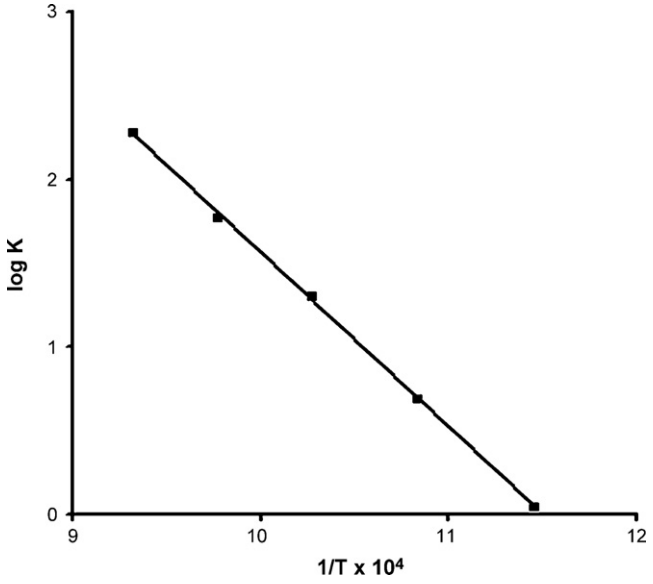


Fig. 6. The Arrhenius plot of the rate constant (K) values calculated by utilizing oxygen diffusion depth (X_{ODZ}) values presented in Fig. 5.

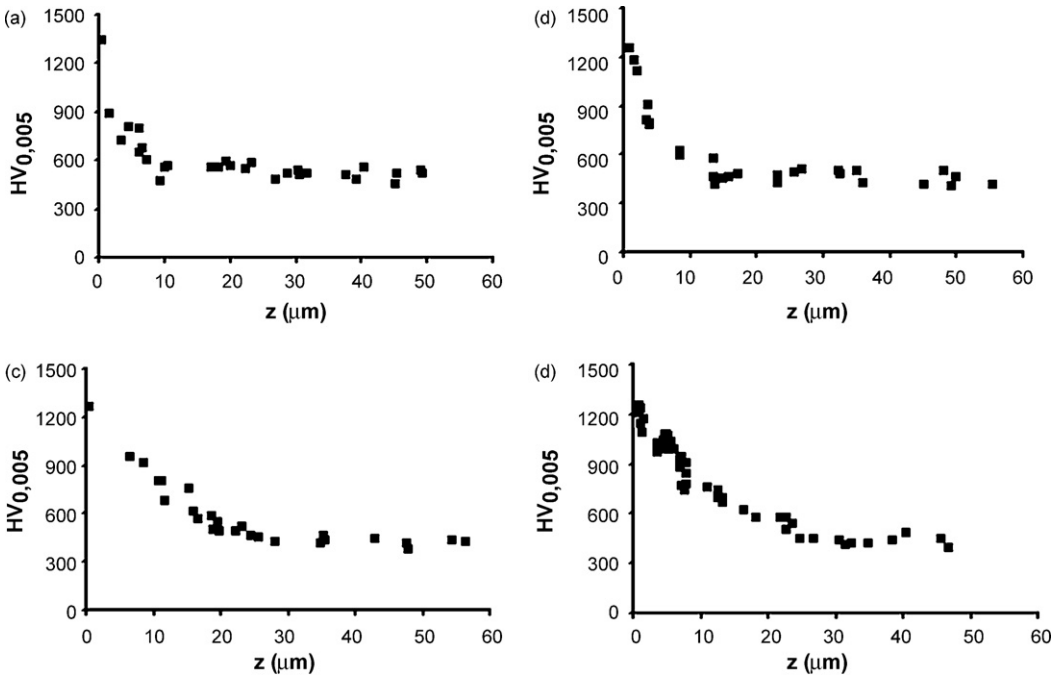


Fig. 7. Hardness (HV)–depth (X) profiles of the samples oxidised at (a) 600 °C for 60 h, (b) 600 °C for 72 h, (c) 650 °C for 60 h and (d) 650 °C for 72 h.

the samples oxidised for 60 and 72 h. "Hardness (HV)–depth (X)" profiles presented in Fig. 7 imply more than two-fold increase in hardness beneath the OL. Gradual decrease of hardness from surface to the core is closely related to oxygen concentration across the ODZ [25–28].

The variation of HV with X can be empirically expressed as

$$HV_X = HV_s \exp(-AX), \tag{3}$$

where HV_X is the hardness at depth X, HV_s is the surface hardness (at $X = 0$) and A is the constant that gives the steepness of the "HV–X" profile. "HV_s" and "A" values of Eq. (3) can be calculated from "ln HV_X vs. X" graphs as presented in Table 1.

It has been reported that "HV–X" profiles obtained through hardness measurements are successfully used to investigate the diffusion kinetics by assuming the correlation between the hardness and the concentration of the dissolved oxygen [3,24–27].

$$\frac{HV_X - HV_s}{HV_0 - HV_s} = \mathrm{erf}\left(\frac{X}{2\sqrt{Dt}}\right), \tag{4}$$

where D is diffusion coefficient and HV_0 is the hardness value at semi-infinite distance from the surface. If no microstructural change takes place during oxidation and/or cooling, HV_0 corresponds to the core hardness. In the present study, HV_0 values were measured as 459 and 438 under the indentation load of 5 g for the samples oxidised at 600 and 650 °C, respectively. After calculation of D, diffusion activation energy (Q_{diff}) can be obtained from [3]:

$$D = D_0\, e^{(-Q_{diff}/RT)}, \tag{5}$$

where D_0 is the diffusion constant.

The solution of Eq. (4) by utilising the HV_X values calculated from Eq. (3) yielded the D values listed in Table 2. Since the D values of two different temperatures (600 and 650 °C) are known for a certain t, Q_{diff} can be calculated from Eq. (5) as 204 and 200 kJ/mol for oxidation times of 60 and 72 h, respectively. Similarly, Frangini et al. [20] reported Q_{diff} value of 200 kJ/mol for a Ti–6Al–4V alloy oxidised at 600–700 °C. It should be noted that, Q_{diff} value calculated by hardness measurements is equal to the Q_{ODZ} value calculated in Section 3.3. For further analysis $Q_{diff} = Q_{ODZ}$ is taken as 202 kJ/mol.

3.4. Temperature-compensated time parameter approach

Since the extent of oxidation and oxygen diffusion is merely related with the exposure time at a given temperature in this study, an attempt has been made to plot the variation of the measured response with respect to a single parameter that combines oxidation temperature and time. Temperature-compensated time parameter has the unit of time, denoted as θ [29]:

$$\theta = t \exp\left(-\frac{Q}{RT}\right) \tag{6}$$

Fig. 8 presents the logarithmic plot of the variation of weight gain and ODZ depth given in Figs. 1 and 5 with respect to θ, which was calculated by inserting the relevant Q values into Eq. (6). Linear equations associated with the regression lines of Fig. 8 are rewritten below in the form of power functions:

$$\frac{\Delta W}{A} = 10^7 \times \theta^{0.5} \tag{7}$$

Table 1
The constants (HV_s and A) of Eq. (3)

Oxidation temperature (°C)	Oxidation time (h)	HV_s	A
600	60	1183	0.0863
	72	1300	0.0943
650	60	1329	0.0457
	72	1254	0.0439

Table 2
Diffusion coefficients (D) and oxygen diffusion activation energy (Q_{diff}) values calculated from Eqs. (4) and (5)

Oxidation temperature (°C)	Oxidation time (h)	D ($\times 10^{-12}$ cm^2/s)	Q_{diff} (kJ/mol)
600	60	0.88 ± 0.04	200
650	60	3.88 ± 0.29	
600	72	0.68 ± 0.06	204
650	72	3.12 ± 0.33	

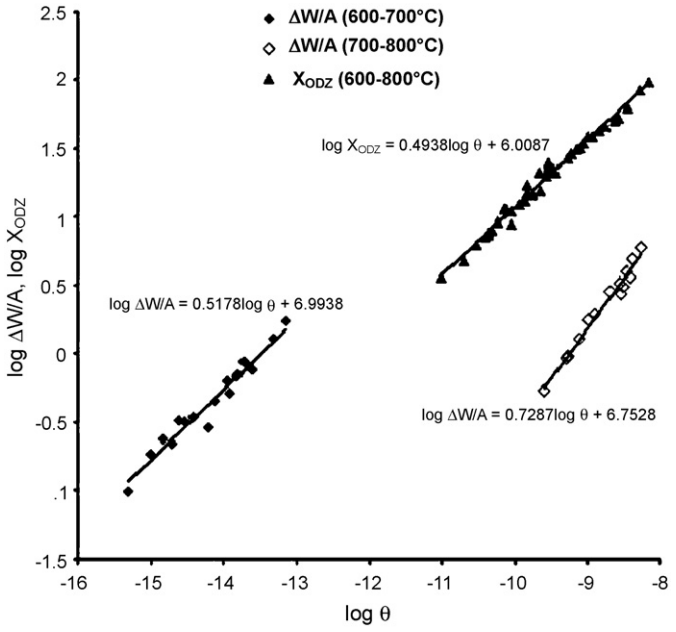


Fig. 8. Logarithmic plot of $\Delta W/A$ and X_{ODZ} experimental data with respect to temperature-compensated time parameter (θ).

$$\frac{\Delta W}{A} = 10^{6.8} \times \theta^{0.73} \quad (8)$$

$$X_{ODZ} = 10^{6} \times \theta^{0.5} \quad (9)$$

These equations stand for the regression analysis of the experimental data with respect to temperature-compensated time parameter and give a possibility for prediction of the extent of oxidation and oxygen diffusion for the time period at a certain oxidation temperature or vice versa. It should be noted that, dominant oxidation mechanism must be considered when calculating the weight gain of Ti–6Al–4V alloy during oxidation. Thus, Eqs. (7) and (8) are valid for parabolic and linear stages of oxidation, respectively. The depth of oxygen diffusion zone can be obtained from Eq. (9), which followed parabolic kinetics over the studied temperature range (600–800 °C).

4. Conclusions

The main conclusions that can be drawn in consequence of this study concerning with the air oxidation behaviour of Ti–6Al–4V alloy at temperatures between 600 and 800 °C for exposure times less than 72 h is as follows:

(1) Oxidation rate obeyed parabolic kinetics between 600 and 700 °C. Parabolic oxidation provided dense and adherent OL on the substrate by maintaining Q_{ox} value of 276 kJ/mol.
(2) Oxidation rate fitted linear kinetics above 700 °C and yielded Q_{ox} value of 191 kJ/mol, by forming porous and crumbly OL over the substrate.
(3) Diffusion of oxygen into the substrate beyond the OL followed parabolic kinetics between 600 and 800 °C by giving Q_{diff} value of 202 kJ/mol.
(4) The parameters of oxidation (temperature and time) can be merged by a temperature-compensated time parameter, which was specified by Eqs. (7)–(9) for prediction of the extent of oxidation and oxygen diffusion.

Acknowledgements

This study was funded by the State Planning Organization of Turkey through the “Advanced Technologies in Engineering” project. The comments of Professor E.S. Kayali on evaluation of experimental results are highly appreciated.

References

[1] R. Boyer, G. Welsch, E.W. Collings, Materials Properties Handbook: Titanium Alloys, ASM International, Materials Park, OH, 1994.
[2] C. Leyens, M. Peters, Titanium and Titanium Alloys: Fundamentals and Applications, Wiley-VCH, Weinheim, 2003.
[3] P. Kofstad, High Temperature Corrosion, Elsevier Applied Science, Essex, 1988.
[4] E. Gryorgy, A.P. Del Pino, G. Sauthier, A. Figuares, F. Alsina, J. Pascual, J. Phys. D: Appl. Phys. 40 (2007) 5246–5251.
[5] I. Gurrappa, J. Alloys Compd. 389 (2005) 190–197.
[6] S. Becker, A. Rahmel, M. Schorr, M. Schutze, Oxid. Met. 38 (2004) 425–464.
[7] Y. Shida, H. Anada, Oxid. Met. 45 (1996) 197–219.
[8] H. Jiang, M. Hirohasi, H. Imanari, Scripta Mater. 46 (2002) 639–643.
[9] A.K. Mishra, J.A. Davidson, R.A. Poggie, P. Kovacs, T.J. FitzGerald, in: S.A. Brown, J.E. Lemons (Eds.), Medical Applications of Titanium and its Alloys, ASTM, Philadelphia, 1996, pp. 96–112.
[10] H. Dong, T. Bell, Wear 238 (2000) 131–137.
[11] J. Komotori, B.J. Lee, H. Dong, P.A. Dearnley, Wear 251 (2001) 1239–1249.
[12] C. Boettcher, T. Bell, H. Dong, Met. Trans. A 33 (2002) 1201–1211.
[13] A. Bloyce, P.Y. Qi, H. Dong, T. Bell, Surf. Coat. Technol. 107 (1998) 125–132.
[14] P.Y. Qi, X.Y. Li, H. Dong, T. Bell, Mater. Sci. Eng. A 326 (2002) 330–342.
[15] H. Dong, X.Y. Li, Mater. Sci. Eng. A 280 (2000) 303–310.
[16] P.A. Dearnley, K.L. Dahm, H. Cimenoglu, Wear 256 (2004) 469–479.
[17] H. Guleryuz, H. Cimenoglu, Biomaterials 25 (2004) 3325–3333.
[18] H. Guleryuz, Surface Treatment of Ti–6Al–4V Alloy by Thermal Oxidation, M.Sc. Thesis, Istanbul Technical University, Turkey, 2003.
[19] H. Guleryuz, H. Cimenoglu, Surf. Coat. Technol. 192 (2005) 164–170.
[20] S. Frangini, A. Mignone, F. De Riccardis, J. Mater. Sci. 29 (1994) 714–720.
[21] H.L. Du, P.K. Datta, D.B. Lewis, J.S. Burnell-Gray, Corros. Sci. 36 (1994) 631–642.
[22] F. Borgioli, E. Galvanetto, F. Iozzelli, G. Pradelli, Mater. Lett. 59 (2005) 2159–2162.
[23] C. Yu, S. Zhu, D. Wei, F. Wang, Surf. Coat. Technol. 201 (2007) 7530–7537.
[24] C. Coddet, A.M. Chaze, G. Beranger, J. Mater. Sci. 22 (1987) 2969–2974.
[25] J.E. Shamblen, T.K. Redden, in: R.I. Jaffee, N.E. Promisel (Eds.), Air Contamination and Embrittlement of Titanium Alloys, The Science, Technology and Application of Titanium, Pergamon Press, London, 1968, pp. 199–208.
[26] J. Unnam, R.N. Shenoy, R.J. Clark, Oxid. Met. 26 (1986) 231–252.
[27] M. Gobel, V.A.C. Haannappel, M.F. Stroosnijder, Oxid. Met. 55 (2001) 137–151.
[28] A.M. Chaze, C. Coddet, J. Mater. Sci. 22 (1987) 1206–1214.
[29] R.W. Hertzberg, Deformation and Fracture Mechanics of Engineering Materials, Wiley & Sons, New York, 1996.